Inverse Functions: Practice Test

Name: _____ Date: _____

5 problems | About 25 minutes | Four-function calculator allowed

Show your work. Write your own explanations and include units in context.

1.

On a fixed-scale trail map, $M(d)$ gives the map distance in centimeters for an actual trail distance of d kilometers. Suppose

$$M^{-1}(7.5) = 18$$

Write a complete-sentence interpretation of this statement. Identify the quantity and units represented by each number.

Answer

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2.

Let *f* be the function below.

$$f(x) = \frac{3x - 8}{2}$$

Find an equation for *g*, the inverse of *f*. Show your algebra.

Work

***g*(*x*) =** _____

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3.

The table gives selected values of a one-to-one function h.

x	$h(x)$
-4	9
3	-2
0	-4
-2	6
9	0
6	3

(a) Find $h^{-1}(-4)$. Explain how the table supports your answer.

Answer

(b) Use the row where $x = -2$ to write the corresponding ordered pair on the graph of h^{-1}. Explain your reasoning.

Answer

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4.

A portable battery initially stores 18 watt-hours of energy. While charging, its stored energy increases by 1.5 watt-hours per minute until it reaches 90.

$$E(t) = 18 + 1.5t$$

Here t is minutes since charging began and $E(t)$ is energy in watt-hours.

(a) Find a formula for E^{-1}. Show your algebra.

Work

(b) For E^{-1}, identify the input and output quantities, including units.

Answer

(c) How long does it take to fully charge the battery? Show your work.

Work

(d) At what time has it gained two-thirds of the additional energy needed to become fully charged? Show your work.

Work

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5.

Let r be the function below.

$$r(x) = \sqrt{18 - 3x}$$

(a) Find r^{-1} and state its domain. Show your algebra.

Work

(b) What should each expression below equal if the functions are inverses? Write your answers first. Then verify each with algebra and state the allowed x-values.

$r(r^{-1}(x)) =$ _____

$r^{-1}(r(x)) =$ _____